Arkaprava Saha

Curriculum Vitae

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Education

2014–Present **Bachelor of Technology in Computer Science and Engineering**, *Indian Institute of Technology Delhi*, New Delhi, *Expected Passing Year: 2018*.
CGPA: 8.2/10 (GPA: 3.7/4) till 6th semester
Pursuing a Departmental Specialization in Data Analytics

2012–2014 **High School**, *D.A.V. Model School*, Durgapur, Affiliated to CBSE.
AISSCE (Final Exam) Grade: 95.8% (Within top 5% in school)

2001–2012 **Secondary School**, *St. Xavier's School*, Durgapur, Affiliated to CISCE.
ICSE (Final Exam) Grade: 95.43% (School Rank 2 out of 165)

Research Interests

- Machine Learning (Data Analytics)
- Theoretical Computer Science
- Programming Languages
- Cryptography

Scholastic Achievements

June 2014 **JEE Advanced (IIT-JEE)**.
Secured **All India Rank 98** out of around 1.3 million students and **State Rank 1**

June 2014 **West Bengal Joint Entrance Examination (WBJEE)**.
Secured **State Rank 6**

December 2012 **Regional Mathematical Olympiad (RMO)**, *Indian Statistical Institute (ISI) Kolkata*, West Bengal, India.
Qualified for the national level

February 2013 **Kishore Vaigyanik Protsahan Yojana (KVPY)**, *Department of Science and Technology (DST)*, Government of India.
Selected as a **Fellow** with **All India Rank 56**

September 2010 **National Talent Search Examination (NTSE)**, *National Council of Educational Research and Training (NCERT)*, Government of India.
Selected as a **Scholar**

Courses Taken

Computer Science Introduction to Computer Science, Data Structures and Algorithms, Discrete Mathematical Structures, Digital Logic and System Design, Computer Architecture, Programming Languages, Design Practices, Computer Networks, Analysis and Design of Algorithms, Principles of Artificial Intelligence, Operating Systems, Introduction to Automata and the Theory of Computation, Introduction to Parallel and Distributed Programming, Introduction to Database Management Systems, Machine Learning (**Graduate Level**), Logic For Computer Science, Special Topics in Machine Learning (Data Mining) (**Graduate Level**), Cryptography (Online Course by **Coursera**)

Basic Sciences Calculus, Linear Algebra and Differential Equations, Introduction to Probability Theory and Stochastic Processes, Electromagnetic Waves and Quantum Mechanics, Principles of Electronic Materials, Introduction to Chemistry, Introduction to Biology for Engineers, Physics and Chemistry Laboratories

Others Introduction to Electrical Engineering, Signals and Systems, Engineering Mechanics, Engineering Visualization and Communication, Product Realization by Manufacturing, Microeconomics, Fundamentals of Language Sciences, Introduction to Sociology, Selected Topics in Linguistics (Morphology)

Technical Skills

Programming Languages C, C++, Java, Python, SQL, SML, VHDL

IDEs Eclipse, BlueJ, MATLAB, MySQL, Android Studio, Xilinx ISE

Web Development HTML, CSS, JavaScript

B. Tech. Project

Random Walks on Bloom Graphs

Advisor **Prof. Maya Ramanath**

Description This project aims to implement random walks on large bloom graphs, whose adjacency lists are stored as lists of **bloom filters**. This will help to answer reachability queries on large graphs like those of social networks with a much smaller memory footprint and a much lower update time than it was possible before. I am **currently working** on this project.

Challenge Bloom filters typically have non-zero probability of **false positives** (predicting that an element is present in the set when it is actually not) and hence the **accuracy** of the random walk generated is a major issue to deal with. Moreover, the space-efficient storage of large graphs and the time-efficient generation of random walks on such graphs are some other challenging tasks.

Role My role in this project is to implement the more efficient **dynamic partition bloom filters** (with sampling) and random walks on both static and dynamic bloom graphs. Also, such walks have to be made as accurate as possible.

Academic Projects

Symmetric Decision Diagrams (SDDs)

Advisor **Prof. Kolin Paul**

Description SDDs are an extension to Binary Decision Diagrams (BDDs) that use symmetry in a boolean function to optimize the number of nodes needed for the function.

Challenge Unlike BDDs, a node in an SDD can correspond to multiple variables. Hence extending boolean operators to SDDs is pretty difficult.

Role My role in the project was to code an implementation of SDDs along with their associated operations (AND, OR, NOT) and creating an SDD from a BDD after obtaining symmetry information from the BDD.

Programming Language Compilers

Advisor **Prof. S. Arun Kumar**

Description This project was about developing a compiler (coded in SML) for a dummy programming language PL. This involved 5 steps: **scanning**, **parsing**, supporting arbitrarily large integers, **machine code generation** and executing that code.

Challenge Parsing the **tokens** from scratch to create a concrete parse tree and then an **abstract syntax tree** was the main challenge. Another difficult task was the generation of machine code from the abstract syntax tree.

Role My role was to code the compiler to generate the **output** of a program in the language PL and to print **error messages** if any, mentioning the error location.

Search Engines

Advisor **Prof. Amitabha Bagchi**

Description This project was about designing a mini search engine in Java that can return results for simple search as well as **AND, OR and phrase search** queries displayed in descending order of a given **relevance** function.

Challenge The main difficulty was to implement suitable data structures to represent the page index and the **inverted page index** for answering all possible queries.

Role My role was to code the search engine using suitable data structures in Java classes, most notably using **AVL Trees** to store the page-location pairs for a given word and a **hash table** to store the entries of all words.

Web Portal for TV Shows

Advisor **Prof. Maya Ramanath**

Description This project involved creating a Web portal for the TV Show **Saturday Night Live** with an SQL database at the back end for storing data (obtained from **Kaggle)** about the show. Users could ask queries related to seasons/episodes.

Challenge Since the data obtained was huge, storing it in a database in a space-efficient manner for the time-efficient answering of queries was a major challenge.

Role My role was to load the database with tables representing the data, program SQL statements for the various possible user queries and create a user interface at the front end for the users to ask queries.

Industrial Internship

Android Client for GWD Tool

Company	**Samsung R&D Institute India, Bangalore**
Duration	May 2017 - July 2017
Mentor	**Preeti Rani Deka**
Description	This was related to the Android Client for the **Samsung Gear Watch Device** (GWD) that displays the selected watch face (dial). Earlier only a few specific watch faces could be rendered on the phone, and all of that was hard coded.
Challenge	Android Views did not provide support for many of the possible features in a watch face, and hence some tweaks had to be thought of.
Role	My role was to create a generalized parser that can parse any general watch face XML file and render it on the phone.

Extra-Curricular Activities

2016	Involved in the **Microsoft Code.Fun.Do Hackathon** at IIT Delhi
2015	Involved in the **ACM-ICPC** Asia Chennai Regionals
2013	Attended the **Vijyoshi Science Camp** at IISc Bangalore for **KVPY Fellows**
2011	Attended the **Nurturance Camp for NTSE Awardees** at RIE Bhubaneswar
Hobbies	Singing, Listening to Music (mainly Indian), Gaming, Watching Sports

Languages

Bengali	**Mother Tongue**
English	**Academic and Professional**
Hindi	**Basic**

References

- **Maya Ramanath**, *Assistant Professor*, Department of Computer Science and Engineering, IIT Delhi.
  ramanath@cse.iitd.ac.in
- **Vinay Joseph Ribeiro**, *Associate Professor*, Department of Computer Science and Engineering, IIT Delhi.
  vinay@cse.iitd.ac.in
- **Preeti Rani Deka**, *Chief Engineer*, Android PF Team, Samsung R&D Institute India, Bangalore.
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